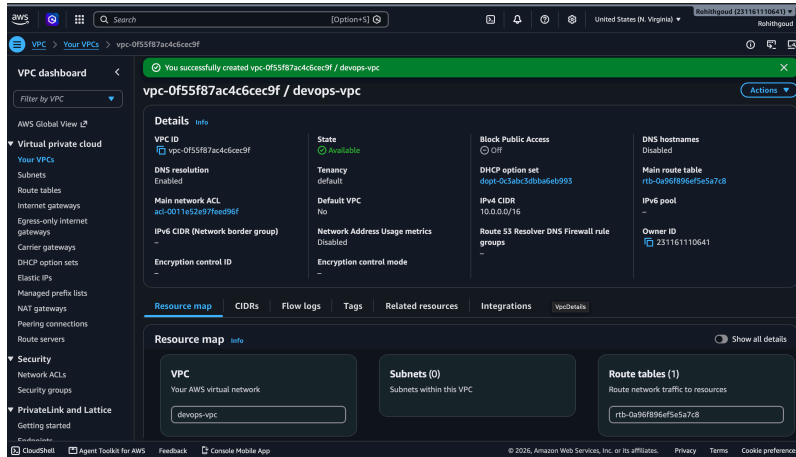
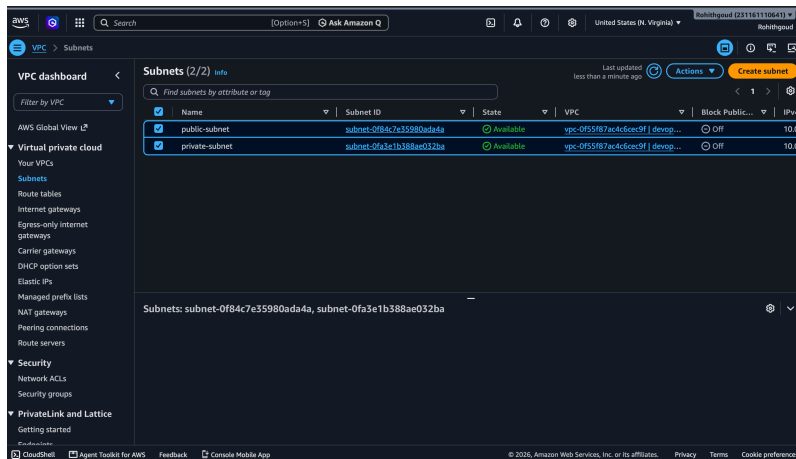


ASG-TASK

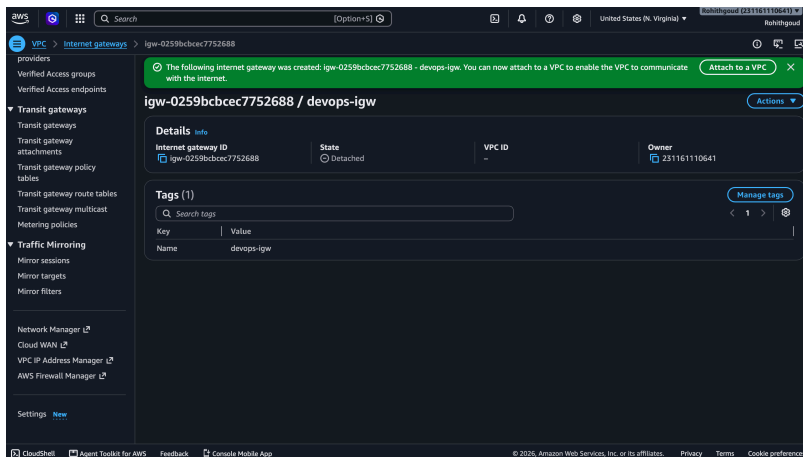
1. Create one VPC in N. Virginia region.



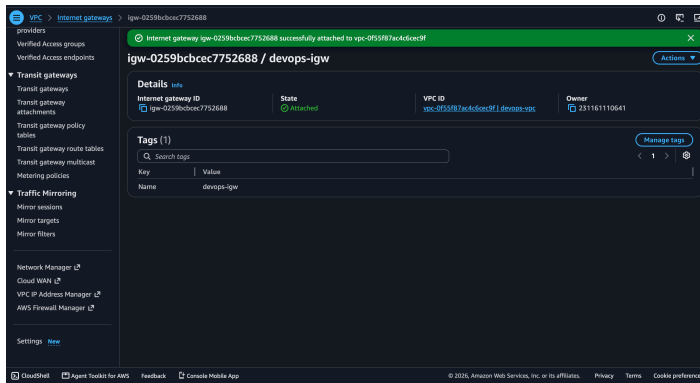
2. Create two subnets: one public subnet and one private subnet.



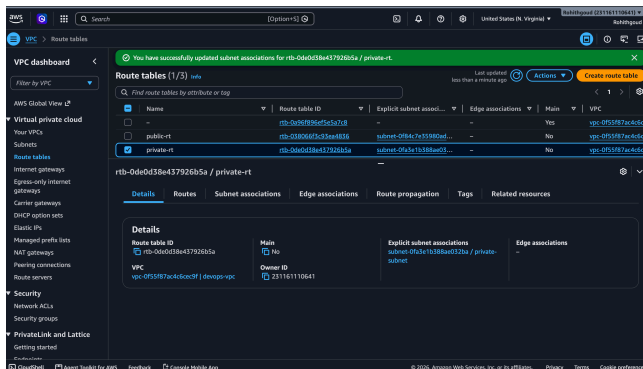
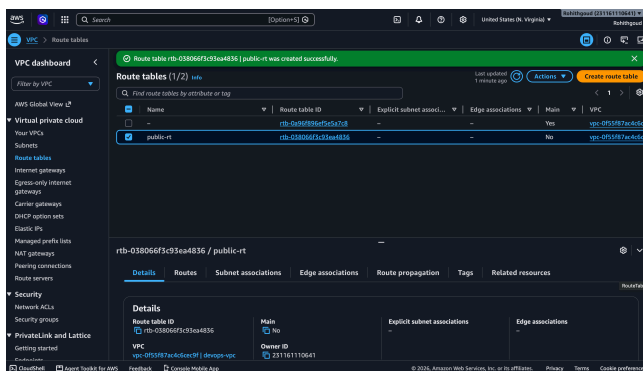
3. Attach an IGW to the VPC.



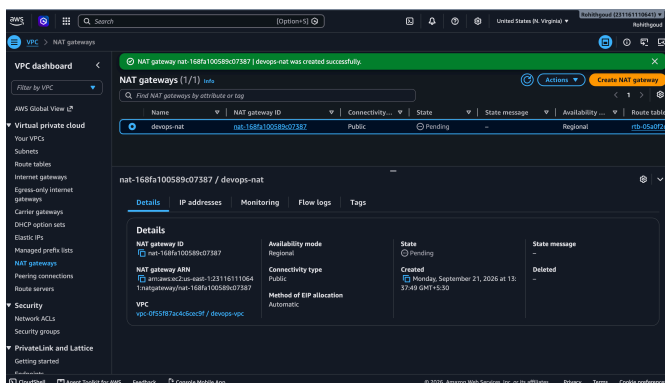
ASG-TASK



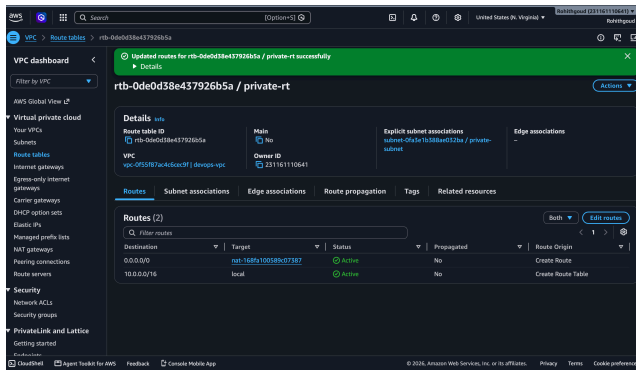
4. Create one public route table (RT) and one private route table.



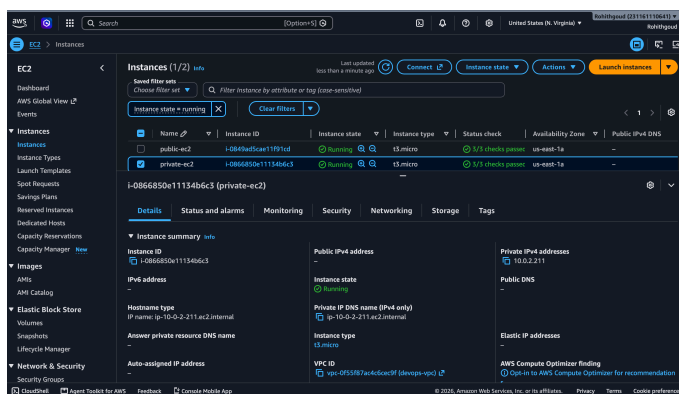
5. Deploy a NAT gateway in the public subnet and attach the NAT gateway to the private subnet.



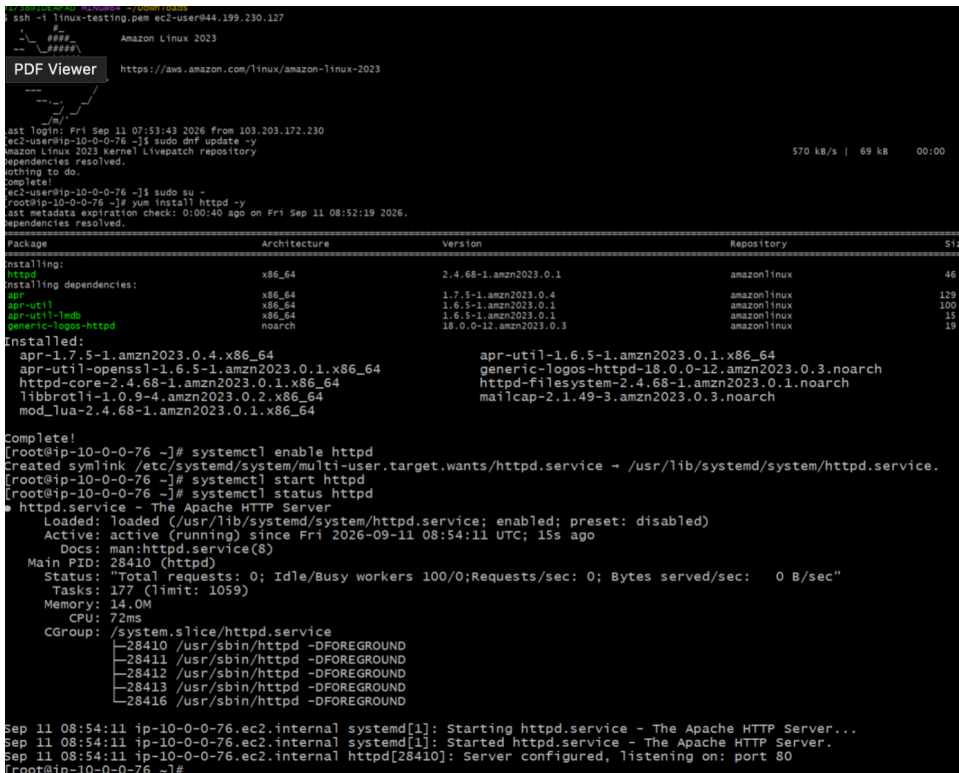
ASG-TASK



6. Create two instances, one in the public subnet and one in the private subnet.



7. Deploy Apache server on both EC2 instances with a sample index.html file.



ASG-TASK

```
root@ip-10-0-0-168 ~# ssh -i /home/ec2-user/linux-amazon-2023 ec2-user@ip-10-0-1.68
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-1-68 ~]$ sudo su -
[root@ip-10-0-1-68 ~]# sudo dnf update -y
Amazon Linux 2023 kernel livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-10-0-1-68 ~]# yum install httpd -y
Last metadata expiration check: 0:00:40 ago on Fri Sep 11 09:00:49 2026.
Package      Architecture      Version            Repository        Size
Installing:  httpd              x86_64            2.4.68-1.amzn2023.0.1  amazonlinux      51 M
Installing dependencies:
httpd-devel    x86_64            1.7.5-1.amzn2023.0.4  amazonlinux      129 M

Installed:
apr-1.6.5-1.amzn2023.0.1.x86_64
apr-util-1.6.5-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
httpd-core-2.4.68-1.amzn2023.0.1.x86_64
libapr1-1.6.5-1.amzn2023.0.1.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
[root@ip-10-0-1-68 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-10-0-1-68 ~]# systemctl status httpd
root@ip-10-0-1-68 ~# systemctl status httpd
httpd.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
Active: active (running) since Fri 2026-09-11 09:02:25 UTC; 16s ago
Docs: man:httd.service(8)
Main PID: 28292 (httpd)
Tasks: Total: 1 (limit: 1059)
      1: /usr/sbin/httpd -DFOREGROUND
      2: /usr/sbin/httpd -DFOREGROUND
      3: /usr/sbin/httpd -DFOREGROUND
      4: /usr/sbin/httpd -DFOREGROUND
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      206: /usr/sbin/httpd -DFOREGROUND
      207: /usr/sbin/http
```

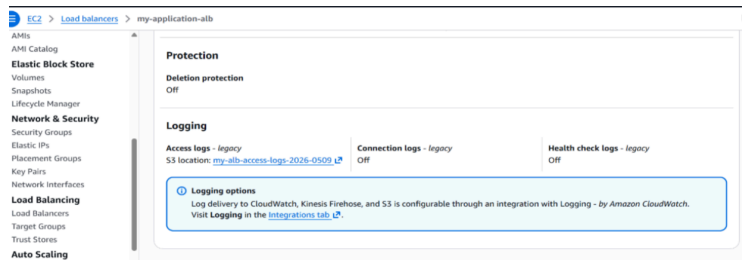
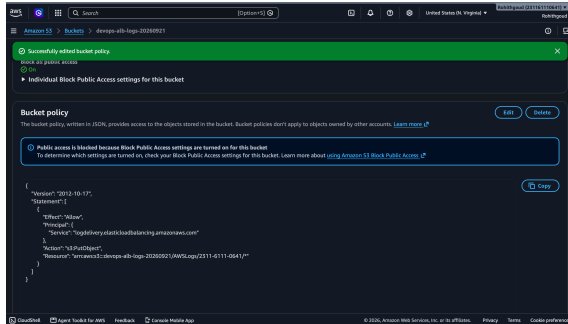
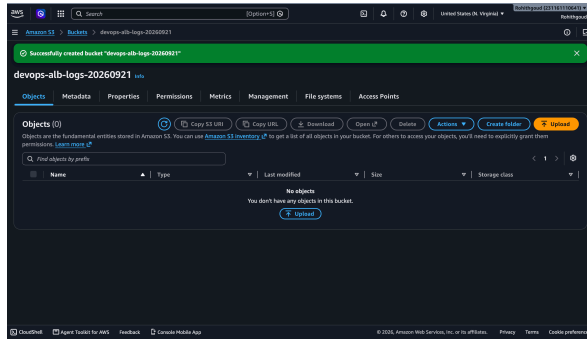
8. Create one application load balancer and attach it to both EC2 instances.

It private works!

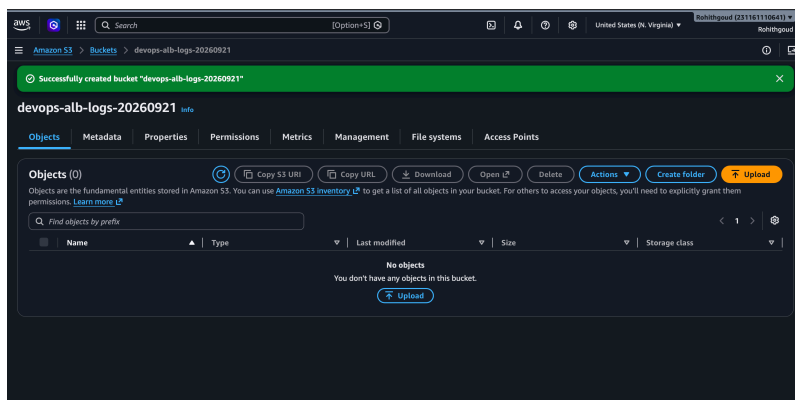
It public works!

ASG-TASK

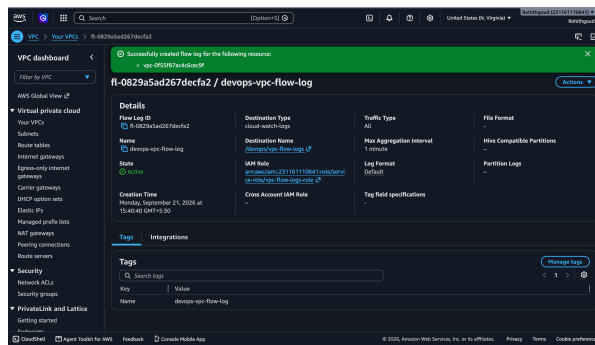
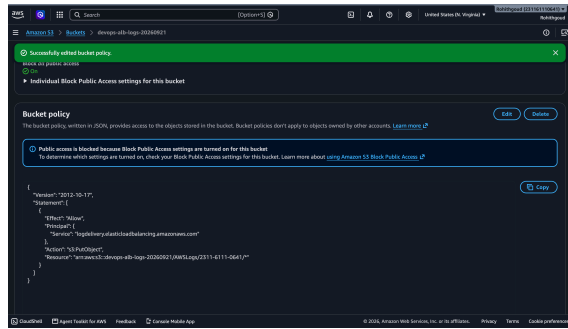
9. Store application load balancer logs in S3.



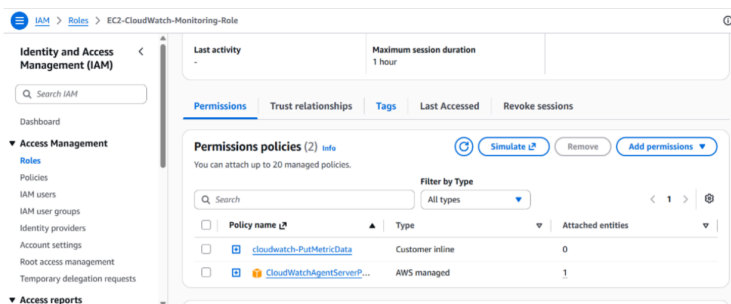
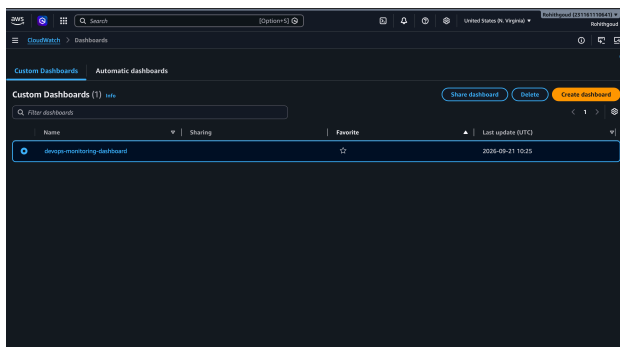
10. Store the VPC flow logs in a CloudWatch log group.



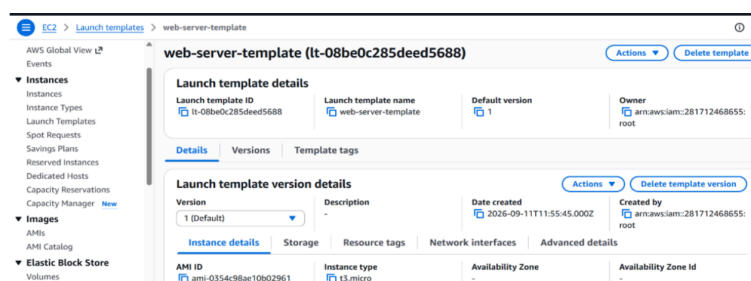
ASG-TASK



11. Create monitoring dashboards to monitor CPU utilization and to monitor the Apache service.



12. If CPU utilization is more than 70%, then it should trigger auto scaling and launch new instance.



ASG-TASK

Hello from Auto Scaling EC2

Apache HTTP Server is running successfully.

This instance was launched by the Auto Scaling Group.